

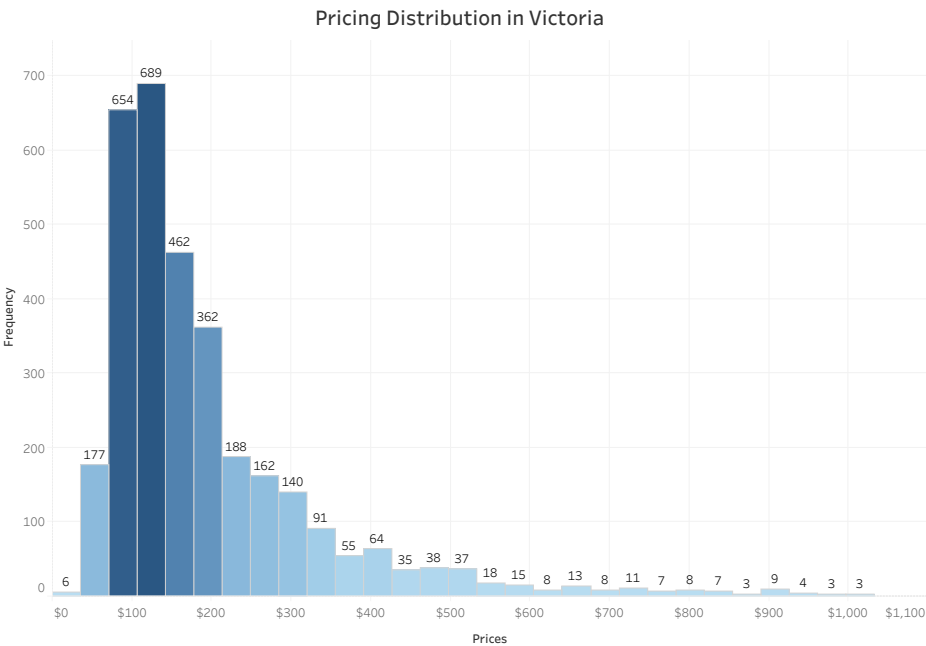
Project Overview	Understanding Pricing Trends by Property T..	Listings and Price by Neighbourhood	Relationship between Price and Number of ..	Pricing based on Clusters	Determining Factors	Conclusions and Recommendations
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Project Overview

Motivation:
With over 5 million hosts and 7.7 million active listings worldwide, Airbnb has become a global leader in the short-term rental market. However, not all listings perform equally—while some generate significant revenue, others struggle to attract bookings. Understanding what drives a listing’s success is essential for both new and existing hosts aiming to optimize their performance.

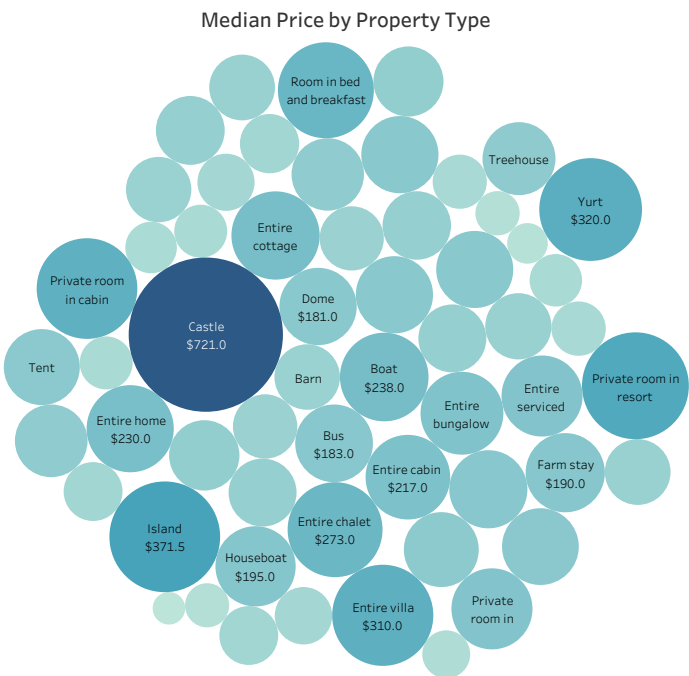
Background on Airbnb:
Founded in 2007 in a San Francisco apartment, Airbnb has revolutionized the hospitality industry by enabling individuals to offer unique accommodations and experiences. By connecting travelers with local communities, Airbnb fosters more authentic and personalized travel experiences.

Objective:
The goal of this project is to identify what makes an Airbnb listing successful by analyzing various performance indicators. This includes examining factors such as location, pricing, guest reviews, host attributes, and available amenities. By uncovering the key drivers behind high-performing listings, we aim to provide actionable insights that can help hosts improve their offerings. Additionally, the project will explore effective pricing strategies to enhance occupancy rates and maximize revenue potential.



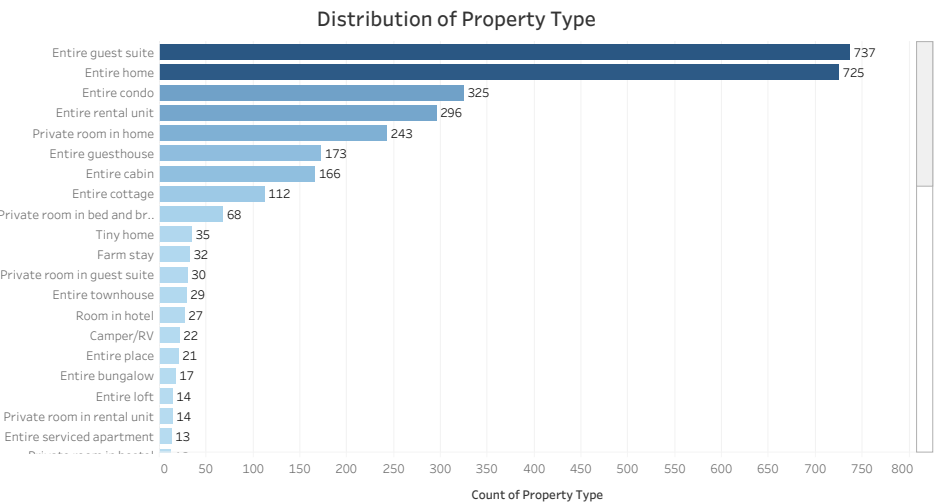
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Understanding Pricing Trends by Property Type



When analyzing the frequency of property types on Airbnb, the **top five most common listings** are **entire guest suites**, **entire homes**, **entire condos**, **entire rental units**, and **private rooms in homes**. These options are widely accessible, affordable, and appeal to a broad range of travelers, making them the backbone of Airbnb's platform.

In contrast, **high-priced property types**—such as **castles**, **private islands**, **resort rooms**, **yurts**, and **entire villas**—are **much less common**. Despite their rarity, these listings offer **unique, premium experiences** that justify their higher rates. They cater to **niche markets** seeking **luxury or novelty**, often featuring exclusive amenities and scenic or remote locations.



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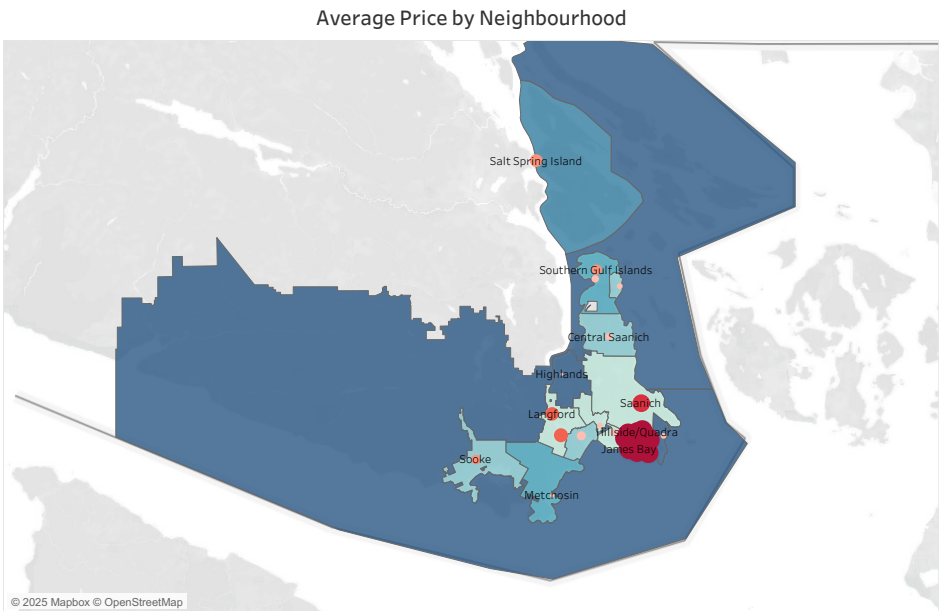
Listings and Price by Neighbourhood

Neighbourhood
All

The **majority of Airbnb listings** are concentrated in **Saanich**, which alone features **over 500 active properties**, making it a central hub for short-term rentals. The **top five neighborhoods with the highest listing volume** are **Saanich, Juan de Fuca, Langford, Salt Spring Island**, and **Downtown**. This concentration suggests that these areas are not only popular with visitors but also **highly competitive for hosts**, requiring strategic pricing and standout features to attract bookings.

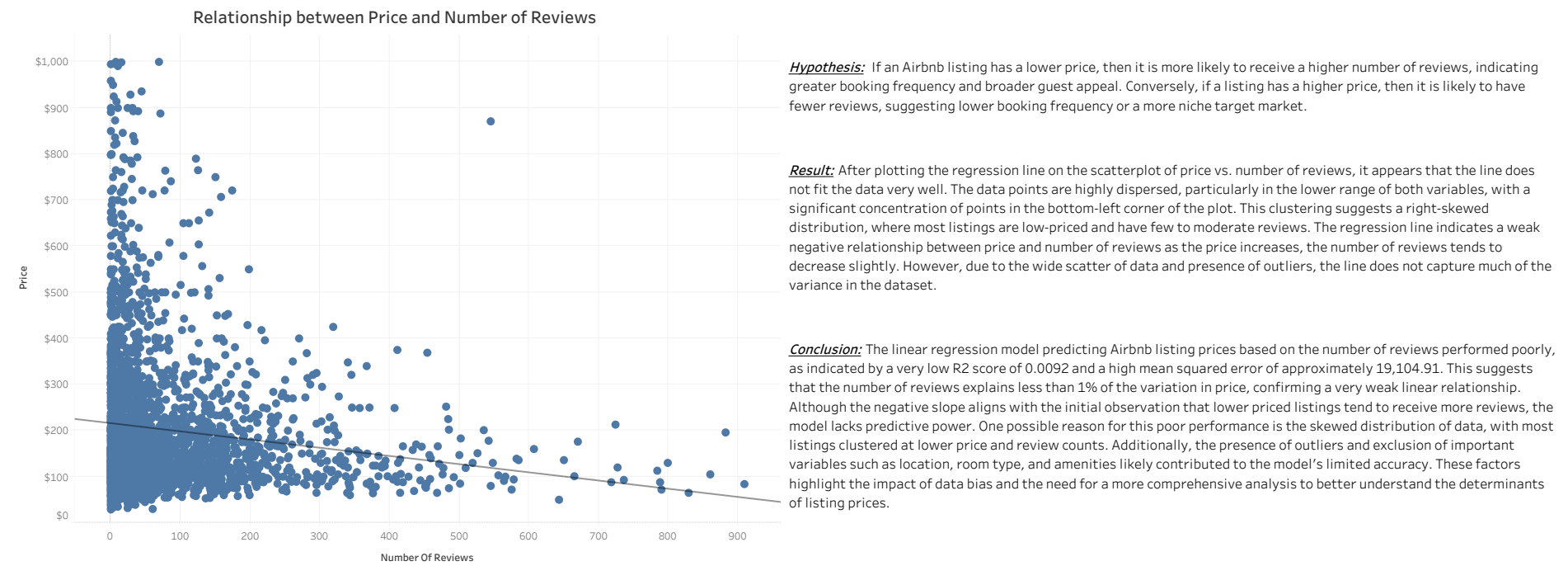
In contrast, the **top five most expensive neighborhoods**, based on **average nightly rates**, are **Juan de Fuca, Southern Gulf Islands, Metchosin, James Bay**, and **Salt Spring Island**. These regions are known for their **scenic views, seclusion, and upscale amenities**, which contribute to their elevated pricing. Although these neighborhoods may have **lower listing density**, their **luxury appeal and exclusivity** attract guests willing to pay a premium for unique or tranquil experiences.

The contrast between **high-volume areas** and **high-priced regions** highlights different host strategies—some aim for **consistent occupancy in competitive markets**, while others target **high-margin, niche traveler segments**. Understanding these dynamics is essential for hosts to position their listings effectively based on location, property features, and guest expectations.



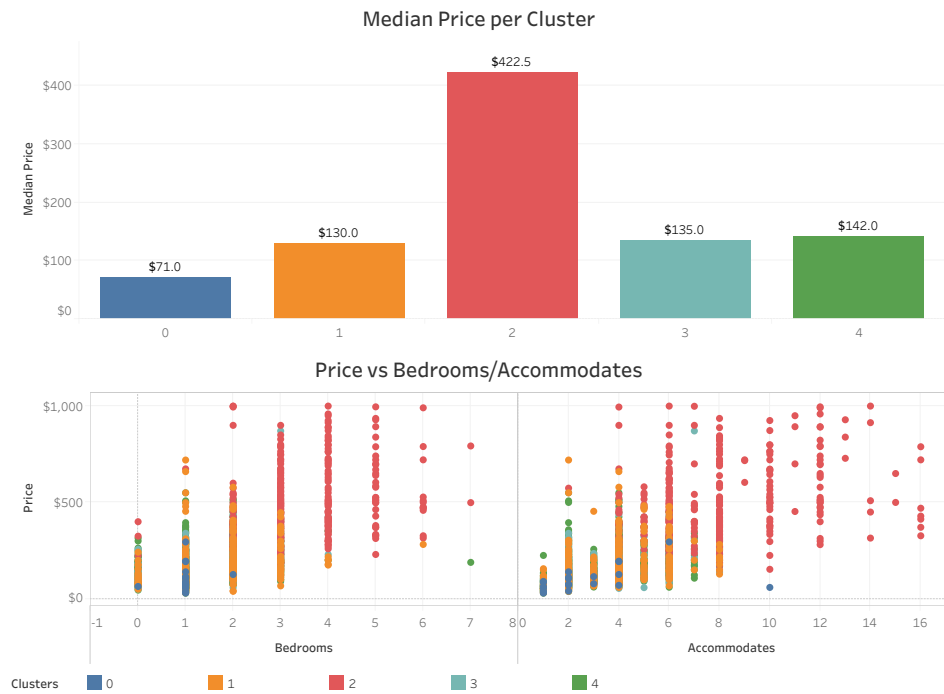
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Relationship between Price and Number of Reviews



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Pricing based on Clusters



To better understand pricing patterns among Airbnb listings, we applied **clustering techniques** to group properties based on shared characteristics such as **price, number of bedrooms, and guest capacity**. The analysis resulted in **five distinct clusters**:

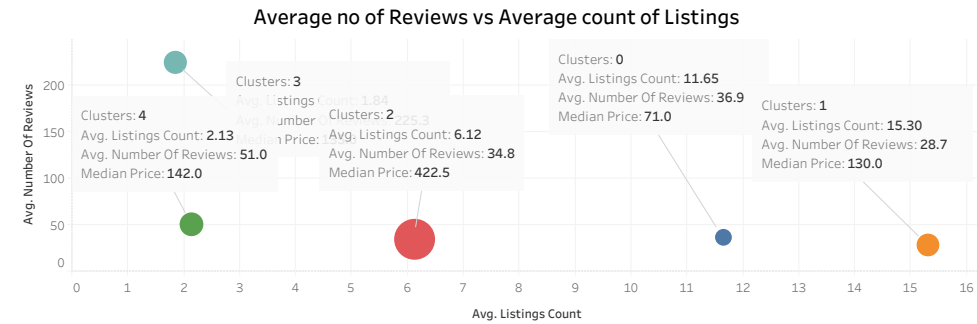
- Cluster 0:** Low-priced listings
- Cluster 1:** Medium-priced listings
- Cluster 2:** High-priced listings
- Cluster 3:** Medium-priced listings
- Cluster 4:** Medium-priced listings

Key Insights:

- **Cluster 2** stands out with the **highest median price of \$422.50**. Listings in this group typically offer **larger accommodations**, featuring **more bedrooms** and the ability to **host more guests**. These are often **premium or luxury properties**, ideal for families or larger groups.
- At the other end of the spectrum, **Cluster 0** represents **budget-friendly listings**, with a **median price of \$71**. These properties are generally **smaller**, with **fewer bedrooms and lower guest capacity**, appealing to **solo travelers or couples** seeking economical stays.
- **Clusters 1, 3, and 4** fall within the **medium price range**, offering a **balanced combination** of affordability and comfort. These listings may include a mix of entire units and private rooms, catering to a **broad customer base**.
- This clustering approach helps us identify how **pricing correlates with listing size and capacity**, providing valuable insights for both **hosts seeking to price competitively** and **travelers searching for value-based options**.

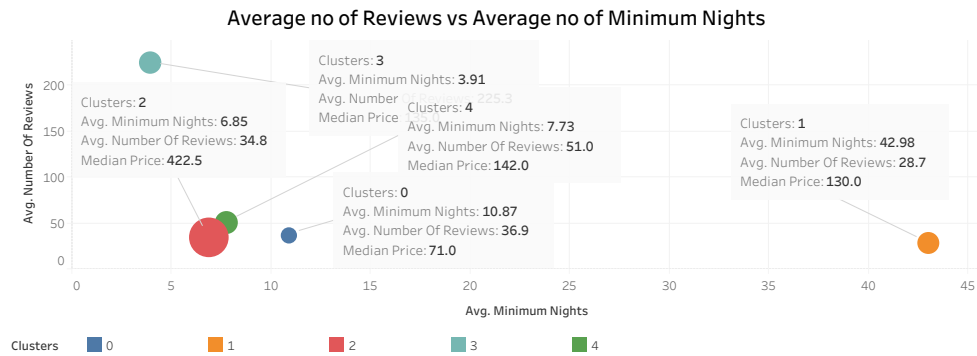
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Determining Factors



Count of Listings appear to influence review activity:

- Cluster 3 stands out with the **highest average number of reviews** per listing (225.3), despite having the **lowest number of listings** (1.84 on average). This suggests that listings in this cluster are **highly popular and frequently booked**, possibly due to their balanced pricing and attractive offerings.
- Cluster 1, in contrast, has the **highest number of listings** (15.3 on average) but the **lowest average number of reviews**. This may indicate **market oversaturation** or **lower guest engagement**, potentially driven by factors like less competitive pricing, weaker amenities, or longer minimum stay requirements.
- Cluster 0 includes the **most expensive listings**. While it doesn't have the highest review count, it maintains a **decent level of engagement**. The higher price point may limit the volume of guests, but the quality of experience could still drive consistent reviews from a smaller base.



Minimum stay requirements appear to influence review activity:

- Cluster 0 requires a **moderate average minimum stay** of 11 nights, balancing exclusivity and accessibility.
- Cluster 1, with the **lowest review count**, requires the **longest minimum stay** at around 43 nights. This extended commitment may discourage short-term travelers, reducing booking frequency and review volume.
- Cluster 2 shows a more balanced behavior—offering **average review numbers** with a **moderately high minimum night requirement**, indicating a **steady but less dynamic listing performance**.

Key Takeaways:

- Clusters with **shorter minimum stays** tend to receive **more reviews**, suggesting better guest turnover and higher engagement.
- Clusters with **very high listing counts** but **low reviews** may need to **reassess their pricing, amenities, or stay policies** to improve performance.
- **High-end listings** (Cluster 0) cater to a niche audience and should focus on **experience quality** rather than volume.

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Conclusions and Recommendations

- Key Conclusions from Airbnb Data (Victoria, BC):
1. The **most common listings** are *entire guest suites, entire homes, condos, rental units, and private rooms*—affordable and appealing to the mass market. **High-priced, rare listings** like *castles, islands, villas, yurts, and resort rooms* offer unique experiences and cater to niche audiences.
 2. **Saanich** leads with over **500 listings**, followed by *Juan de Fuca, Langford, Salt Spring Island, and Downtown*—all representing **high-traffic, competitive areas**.
 3. The **most expensive neighborhoods** (*Juan de Fuca, Southern Gulf Islands, Metchosin, James Bay, and Salt Spring Island*) have **fewer listings** but offer **premium experiences** and higher pricing.
 4. A **weak negative correlation** exists between **price** and **number of reviews**. Listings with **lower prices** tend to receive **more reviews**, but this relationship is not strong or predictive due to **skewed data and outliers**.
 5. Cluster 3 has the fewest listings but highest reviews, Cluster 0 features the most expensive listings with moderate reviews and 11-night stays, Cluster 1 has the most listings but lowest reviews due to long 43-night minimums, while Clusters 2 and 4 offer medium to high pricing with balanced guest engagement.

- Recommendations:
1. **Incorporate Neighborhood Effects:** Pricing strategies should account for neighborhood popularity and guest demand, as location significantly impacts listing performance.
 2. **Prioritize Fewer Listings:** Hosts should focus on optimizing 1–2 listings rather than spreading resources thin, which can lead to higher guest engagement and more reviews.
 3. **Shorten Minimum Stay Requirements:** Reducing the minimum number of nights can increase booking frequency and review volume, especially in competitive markets.

- Future Steps:
1. **Perform Review Sentiment Analysis:** Use natural language processing (NLP) to extract keywords and emotional tone from guest reviews to uncover what drives satisfaction or dissatisfaction.
 2. **Analyze Amenities Through Text Parsing:** Move beyond just room count or guest capacity—examine listing descriptions to identify how specific amenities influence pricing.
 3. **Improve Cluster Balance:** Revisit clustering using alternative algorithms to achieve more meaningful and evenly distributed groupings.

Data Limitations: The dataset is confined to information gathered on December 29, 2024, and does not account for any modifications to listings, prices, or reviews made after that date. Consequently, the data may not accurately reflect the curr..

Distribution of Clusters

